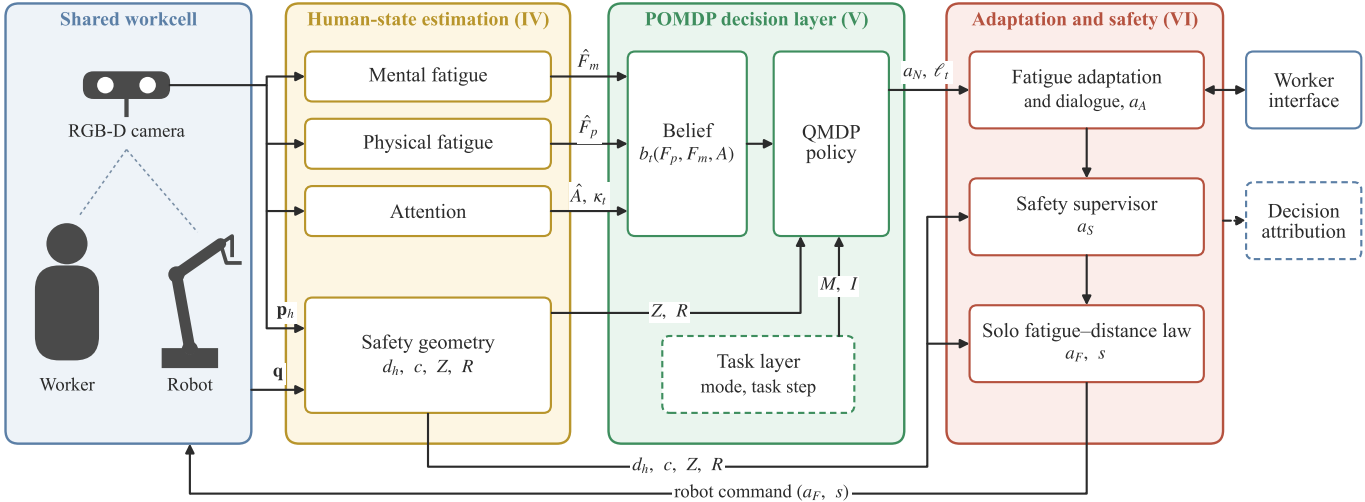
Shared workcell
RGB-D camera
Worker
Robot
Human-state estimation (IV)
Mental fatigue
Physical fatigue
Attention
Safety geometry
$d_h,\ c,\ Z,\ R$
$\mathbf{p}_h$
$\mathbf{q}$
$\hat{F}_m$
$\hat{F}_p$
$\hat{A},\ \kappa_t$
POMDP decision layer (V)
Belief
$b_t(F_p, F_m, A)$
QMDP policy
$Z,\ R$
$M,\ I$
Task layer
mode, task step
$a_N,\ \ell_t$
Adaptation and safety (VI)
Fatigue adaptation and dialogue, a_A
Safety supervisor
a_S
Solo fatigue–distance law
$a_F,\ s$
Worker interface
Decision attribution
$d_h,\ c,\ Z,\ R$
robot command $(a_F,\ s)$